Customer Support Chatbot

By Eng: Ahmed Mohamed Hussein

1. Introduction

This document presents the complete technical documentation for the Customer Support ChatBot system an end-to-end NLP pipeline that classifies customer support messages and routes them to a Large Language Model (LLM) for intelligent, context-aware responses.

The system is built around two core components:

- A Dual RoBERTa Classifier that simultaneously predicts the intent and category of incoming customer messages with near-perfect accuracy.
- A conversational LLM backend powered by LLaMA 3.3 70B (via Groq Cloud) that uses the classifier output to generate professional customer service responses.
- A Streamlit web application providing a clean zero-code interface where users can describe their issues, view predictions, and chat interactively with the AI agent.

1.1 System Overview

The project was designed as a modular pipeline where each component solves one specific problem. The classifier handles intent routing traditionally a rules-based or keyword-matching task with a fine-tuned transformer that understands natural language variation, colloquialisms, typos, and emotional language. The LLM then uses this structured routing context to respond as a professional customer service employee.

	Key Technical Stack
Classifier	RoBERTa-base (HuggingFace Transformers) — fine-tuned twice, once per task
LLM Backend	LLaMA 3.3 70B Versatile via Groq Cloud API
UI Framework	Streamlit
Training	HuggingFace Trainer with class-weighted cross-entropy loss
Hardware	NVIDIA GPU (CUDA-enabled)

2. Dataset

2.1 Source & Description

The dataset used is the Bitext Customer Support LLM Chatbot Training Dataset, published on HuggingFace by Bitext Innovations. It is a professionally curated dataset specifically designed for training and evaluating intent classification and chatbot systems in the customer support domain.

Parameter	Value / Description
Source	HuggingFace Hub — bitext/Bitext-customer-support-llm-chatbot-training-dataset
Total Samples	26,872 utterances
Columns	flags, instruction, category, intent, response
Input Column	instruction — the raw customer utterance
Target (task 1)	category — 11 high-level topic categories
Target (task 2)	intent — 27 fine-grained intent labels
Missing Values	0 missing values in instruction, category, or intent
Language	English (with colloquialisms, typos, and profanity variations)

2.2 Label Classes

Category Labels (11 classes)

The 11 categories represent high-level support topics. Each category aggregates multiple related intents.

Parameter	Value / Description
ACCOUNT	Account management, profile changes, login issues
CANCEL	Early termination, cancellation fees, withdrawal
CONTACT	Reaching human agents, customer service contact
DELIVERY	Delivery methods, estimated arrival, tracking
FEEDBACK	Complaints, claims, and product/service feedback
INVOICE	Billing documents, invoice retrieval
ORDER	Order placement, order details, wrong items
PAYMENT	Payment methods, billing issues, double charges
REFUND	Return requests, refund eligibility and status

SHIPPING	Shipping address, shipping options and costs
SUBSCRIPTION	Subscription plans, renewals, and modifications

2.3 Class Distribution

The dataset is significantly imbalanced. ACCOUNT is the most frequent class with 5,986 samples, while CANCEL is the least frequent with only 950 samples — an imbalance ratio of 6.30x. This imbalance was addressed during training using class-weighted cross-entropy loss.

Parameter	Value / Description
ACCOUNT	5,986 (22.3%)
ORDER	3,988 (14.8%)
REFUND	2,992 (11.1%)
CONTACT	1,999 (7.4%)
INVOICE	1,999 (7.4%)
PAYMENT	1,998 (7.4%)
FEEDBACK	1,997 (7.4%)
DELIVERY	1,994 (7.4%)
SHIPPING	1,970 (7.3%)
SUBSCRIPTION	999 (3.7%)
CANCEL	950 (3.5%)

2.4 Data Split

The full dataset was split into three stratified subsets to ensure representative class proportions in each split:

Parameter	Value / Description
Train Set	18,810 samples (70.0%)
Validation Set	4,031 samples (15.0%)
Test Set	4,031 samples (15.0%)
Stratification	Applied — preserves class balance across all splits
Random Seed	42 (reproducible)

3. Exploratory Data Analysis (EDA)

3.1 Label Distribution

The charts below show the sample count and percentage share for each category. The clear dominance of ACCOUNT category and the underrepresentation of CANCEL and SUBSCRIPTION categories motivated the use of class weights during training.

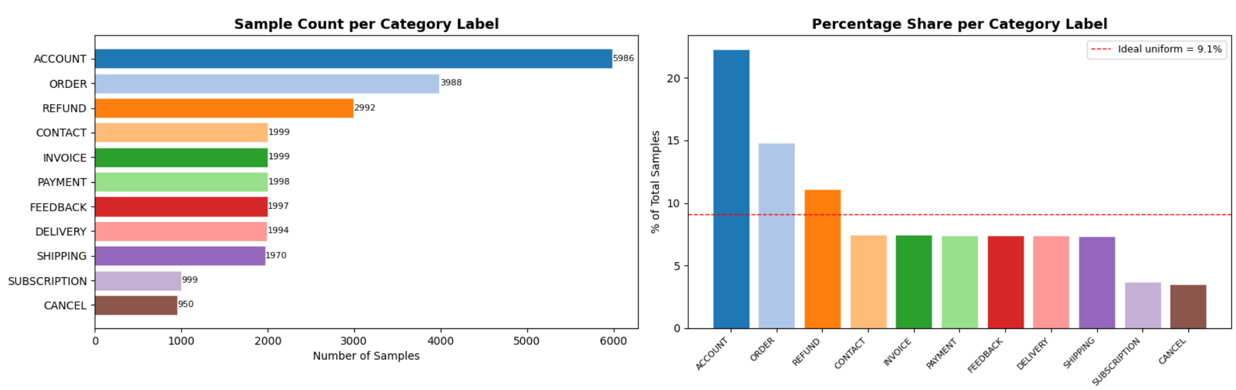


Figure 1: Label distribution — sample count (left) and percentage share (right) per category

3.2 Utterance Length Analysis

Customer utterances in this dataset are extremely short. The distribution of word counts reveals a median of 9 words and a mean of 8.7 words, with a maximum of 16 words. This short-text nature is a key factor in model architecture selection.

Parameter	Value / Description
Mean Word Count	8.69 words
Median Word Count	9.0 words
Std Deviation	2.6 words
Minimum	1 word
Maximum	16 words
25th Percentile	7 words
75th Percentile	11 words

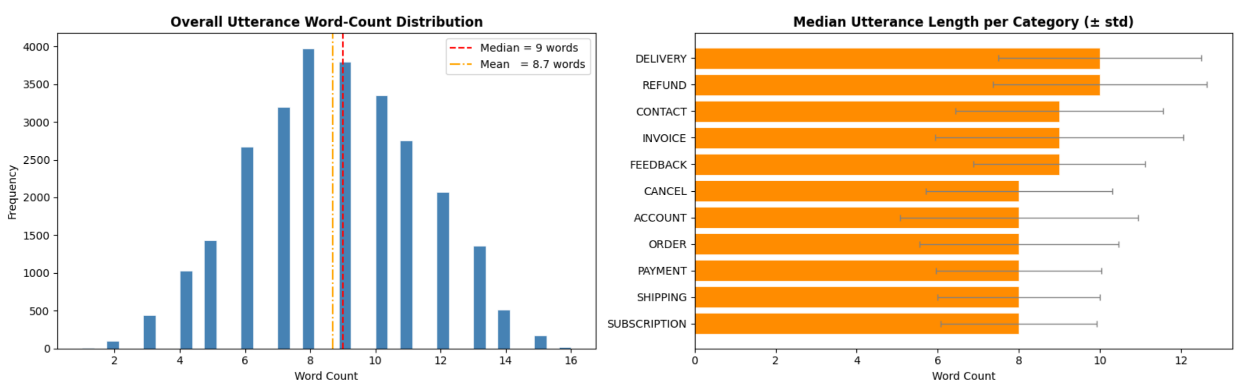


Figure 2: Overall utterance word-count distribution (left) and median length per category with std deviation (right)

A critical observation is that utterance length is essentially uniform across all 11 categories, with median values clustering tightly between 8 and 10 words per class. This confirms that text length alone provides zero discriminatory power for classification, and that semantic meaning must be captured through contextual embeddings.

3.3 Corpus-Wide Word Frequency

After removing standard English stopwords and domain-specific common terms, the top words across the full corpus directly reflect the category labels themselves terms like "account", "order", "payment", and "refund" dominate. This vocabulary-label alignment is expected in a well-constructed intent dataset.

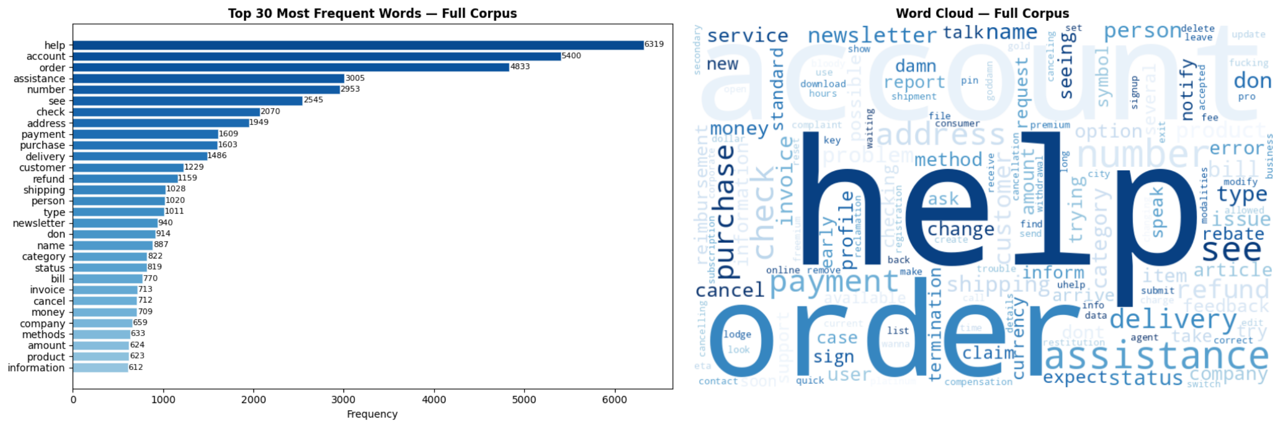


Figure 3: Top 30 most frequent words corpus-wide (left) and word cloud visualization (right)

However, generic support verbs "help", "assistance", "check", "information" appear in the top 10 across virtually every category. This inter-class vocabulary overlap is the primary challenge for simple Bag-of-Words models and validates the use of a transformer-based approach.

3.4 Per-Class Word Clouds

Individual word clouds were generated for each of the 11 categories to visualize the unique vocabulary per class and identify potential data artifacts.

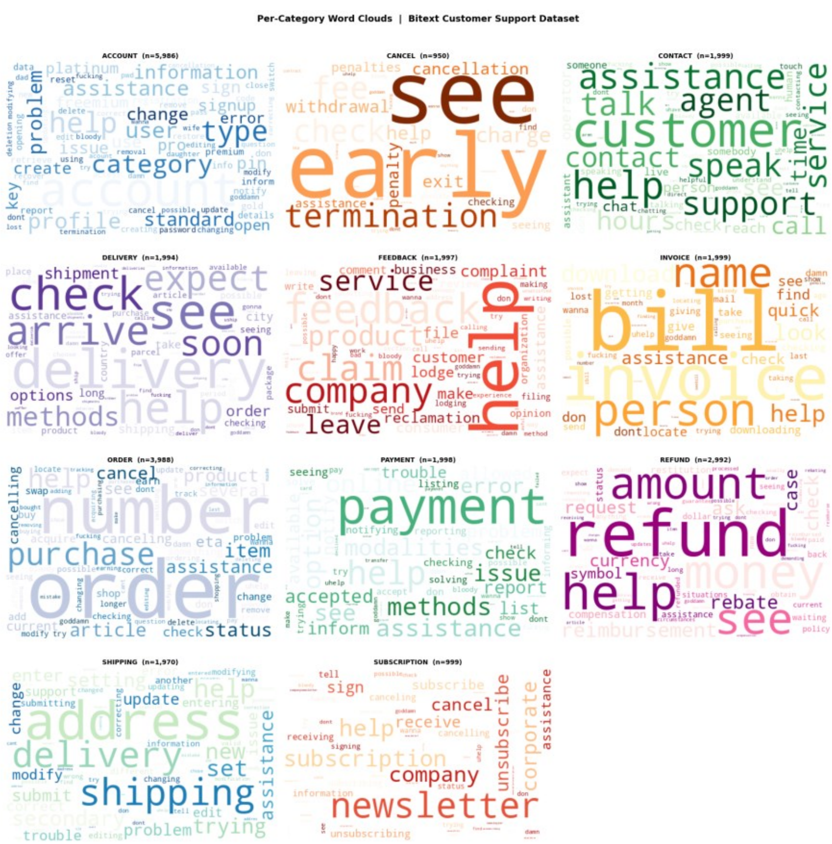


Figure 4: Per-category word clouds — each cloud highlights the most distinctive vocabulary for that support topic

Key observations from the per-class word clouds:

- ACCOUNT and ORDER classes are dominated by their namesake keywords, indicating strong class-specific vocabulary signal.
- SHIPPING and DELIVERY share significant vocabulary overlap, making them the most likely pair for misclassification.
- The INVOICE class showed numeric artifacts (synthetic template IDs), a potential source of overfitting on specific string patterns.
- Profanity tokens appear across multiple classes, reflecting the emotional variation flags in the dataset (colloquial and frustrated utterances).

3.5 Top-10 Words Per Category

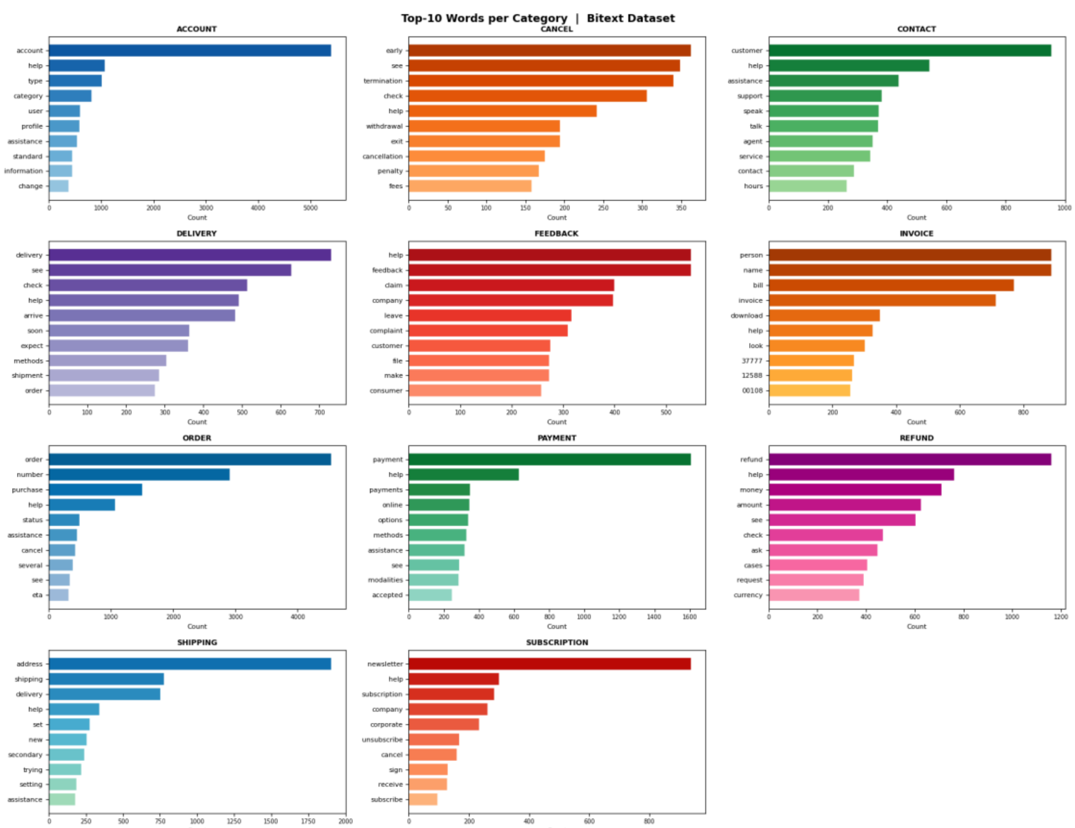


Figure 5: Top-10 most frequent words per category after stopword removal

4. Text Cleaning

Given that RoBERTa is a transformer with its own subword tokenization, heavy text preprocessing is counterproductive it can destroy signals that the model uses for contextual understanding. A minimal cleaning pipeline was designed to preserve the natural linguistic diversity of the dataset while removing only genuinely noisy content.

4.1 Cleaning Pipeline

Parameter	Value / Description
URL Removal	Strips any http:// or www. links (rare in utterances but present in templates)
HTML Tag Removal	Removes any <tag> artifacts from template generation
Whitespace Normalization	Collapses multiple spaces/newlines into a single space and strips leading/trailing whitespace
Preserved	Typos, colloquialisms, punctuation, capitalization, profanity — all intentionally retained

---

5. Tokenization

Tokenization converts the cleaned text into numerical token sequences that can be processed by the transformer model. The RoBERTa tokenizer uses Byte-Pair Encoding (BPE), which splits words into subword units and handles out-of-vocabulary terms robustly.

5.1 Tokenizer Configuration

Parameter	Value / Description
Model	roberta-base
Tokenizer Type	Byte-Pair Encoding (BPE)
Vocabulary Size	50,265 tokens
Max Length	64 tokens (truncation applied if exceeded)
Padding Strategy	Dynamic padding per batch via DataCollatorWithPadding
Special Tokens	<s> (CLS), </s> (SEP), <pad>, <unk>, <mask>

5.2 Max Length Selection

A maximum sequence length of 64 tokens was selected based on the corpus statistics. With utterances averaging 8.7 words and a maximum of 16 words, even the longest utterances tokenize to well under 30 tokens. Using 64 instead of the model's default 512 reduces GPU memory consumption by 8x and speeds up training significantly with zero accuracy penalty.

5.3 Dynamic Padding

Dynamic padding (padding to the longest sequence in each batch rather than to the global max length) was implemented using DataCollatorWithPadding. This further reduces computational waste per batch, particularly important for a short-text corpus where batch lengths vary widely.

5.4 Dataset Sizes After Tokenization

Parameter	Value / Description
Train Dataset	18,810 rows × {input_ids, attention_mask, label}
Val Dataset	4,031 rows × {input_ids, attention_mask, label}
Test Dataset	4,031 rows × {input_ids, attention_mask, label}

6. Model Setup & Configuration

6.1 Base Model: RoBERTa-base

Parameter	Value / Description
Architecture	Transformer Encoder (12 layers, 12 attention heads)
Hidden Size	768 dimensions
Total Parameters	124,654,091
Trainable Params	124,654,091 (full fine-tuning)
Classification Head	Dense(768 → 11 classes) for category; Dense(768 → 27 classes) for intent
Pretrained On	Books Corpus + Wikipedia + CC-News + OpenWebText + Stories

RoBERTa (Robustly Optimized BERT Pretraining Approach) is a transformer encoder model pretrained on a large English corpus using masked language modeling. It was chosen for this task because of its strong performance on short-text classification tasks, large vocabulary (50K tokens) which handles diverse customer language, and its well-established fine-tuning stability.

6.2 Custom Weighted Trainer

The standard HuggingFace Trainer was subclassed to inject class-weighted cross-entropy loss. This addresses the dataset imbalance by penalizing misclassifications of rare classes (CANCEL, SUBSCRIPTION) more heavily than misclassifications of frequent classes (ACCOUNT, ORDER).

Parameter	Value / Description
ACCOUNT	0.4081 (most frequent — lower penalty)
ORDER	0.6125
REFUND	0.8166
CONTACT	1.2223
INVOICE	1.2223
PAYMENT	1.2232
FEEDBACK	1.2232
DELIVERY	1.2249
SHIPPING	1.2400
SUBSCRIPTION	2.4464
CANCEL	2.5676 (least frequent — higher penalty)

6.3 Training Configuration

Parameter	Value / Description
Learning Rate	2e-5 (standard RoBERTa fine-tuning rate)
Batch Size	32 train / 64 eval
Max Epochs	10 (with early stopping, patience=3)
Warmup Steps	200
LR Scheduler	Cosine decay
Weight Decay	0.01 (L2 regularization)
Max Sequence Length	64 tokens
Mixed Precision	FP16 (when CUDA available)
Model Selection Metric	Macro F1 (best model saved)
Checkpointing	Save 2 best checkpoints per task
Seed	42

7. Model Results

7.1 Training Summary

Training converged rapidly due to the clean, well-structured nature of the Bitext dataset. Early stopping triggered after 4 epochs (patience=3), confirming that the model reached its optimal state well before the maximum epoch limit.

Parameter	Value / Description
Total Training Time	296.1 seconds (~5 minutes)
Throughput	635.3 samples/second
Final Training Loss	0.2109

Epochs Completed	4 (early stopping triggered)
Best Epoch	4

7.2 Validation & Test Set Performance

Metric	Val Loss	Val Accuracy	Macro F1	Weighted F1
Validation Set	0.0090	99.88%	0.9988	0.9988
Test Set	0.0017	99.98%	0.9996	0.9998

The test set results show near-perfect classification performance across all metrics. The test loss of 0.0017 is remarkably low, reflecting the model's very high confidence in its predictions. The macro F1 of 0.9996 indicates that performance is consistent even across minority classes.

---

8. Evaluation Metrics

8.1 Per-Class Classification Report

The table below shows the precision, recall, F1-score, and support for each category class on the test set (4,031 samples total).

Class	Precision	Recall	F1-Score	Support
ACCOUNT	1.00	1.00	1.00	898
CANCEL	1.00	1.00	1.00	142
CONTACT	1.00	1.00	1.00	300
DELIVERY	1.00	1.00	1.00	299
FEEDBACK	1.00	1.00	1.00	300
INVOICE	1.00	1.00	1.00	300
ORDER	1.00	1.00	1.00	598

PAYMENT	1.00	1.00	1.00	300
REFUND	1.00	1.00	1.00	449
SHIPPING	1.00	1.00	1.00	295
SUBSCRIPTION	1.00	0.99	1.00	150
Accuracy			1.00	4031
Macro Avg	1.00	1.00	1.00	4031
Weighted Avg	1.00	1.00	1.00	4031

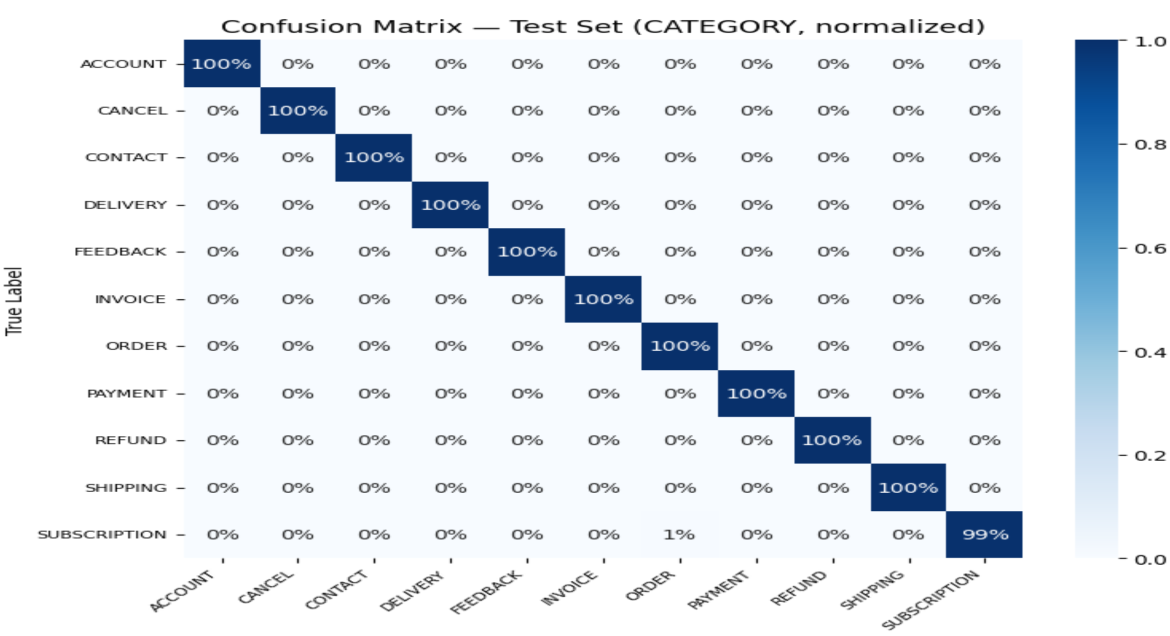
8.2 Error Analysis

With only 1 misclassification out of 4,031 test samples (0.02% error rate), the error analysis is almost trivial for the category model. The single error was a SUBSCRIPTION utterance that was predicted as ORDER — a semantically adjacent pair where "subscription" language can overlap with "order" language in edge cases.

8.3 Confusion Matrix

The normalized confusion matrix below confirms the near-perfect diagonal — all classes achieve 100% recall except SUBSCRIPTION, which shows a single off-diagonal entry (1 sample misclassified as.

Figure 6: Normalized confusion matrix on the test set (CATEGORY model). Diagonal is 100% for 10 out of 11 classes.



9. Training Curves & Analysis

9.1 Loss Curves

The training and validation loss curves are shown below for the 4 completed epochs.

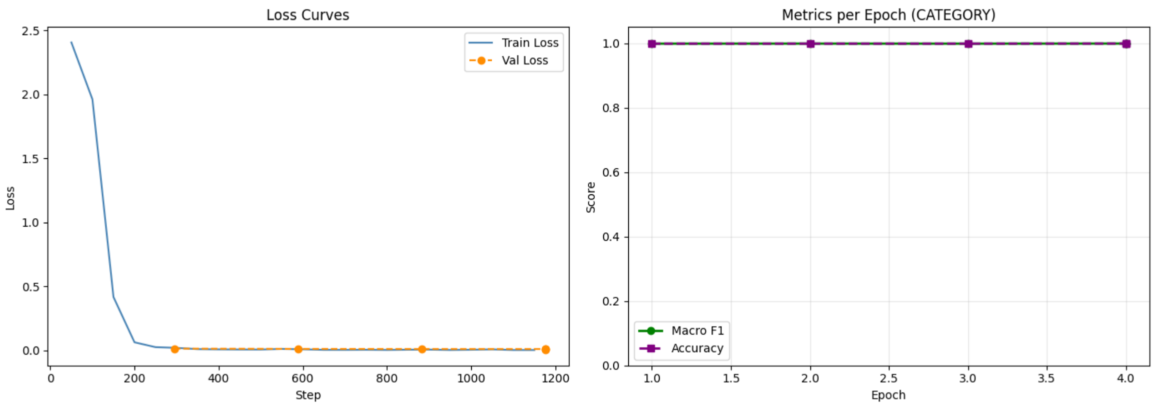


Figure 7: Training loss per step (left) and macro F1 / accuracy per epoch on validation set (right)

Key observations from the loss curves:

- Training loss drops steeply from the first epoch, confirming that the pretrained RoBERTa representations already encode strong semantic features relevant to the task.
- Validation loss closely tracks training loss throughout, with no divergence — this indicates no overfitting occurred despite fine-tuning all 124M parameters.
- Both macro F1 and accuracy plateau above 99.85% from epoch 1 onwards, showing rapid convergence.

9.2 Epoch-Level Curves

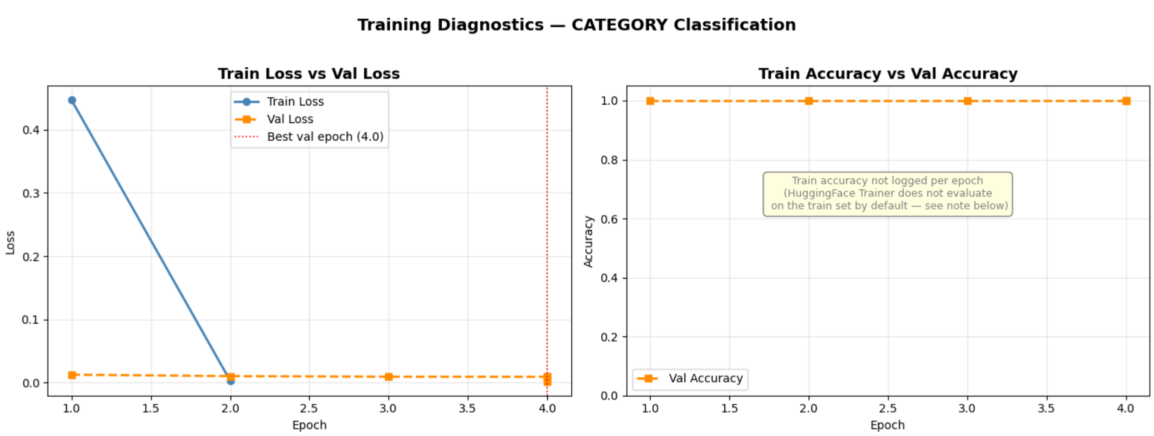


Figure 8: Train loss vs. validation loss per epoch (left) and validation accuracy per epoch (right). Red dotted line marks the best validation epoch.

Key observations from the epoch-level view:

- Train loss and validation loss decrease in parallel, with the validation loss reaching a very low 0.009 by epoch 4. The near-zero gap between the two curves confirms strong generalization.
- Validation accuracy climbs from 99.85% at epoch 1 to 99.98% at epoch 4, with consistent monotonic improvement across all epochs.
- The red dotted line at epoch 4 marks the best checkpoint, which is the model saved for inference.
- Early stopping triggered after epoch 4 (patience=3 with threshold=0.001), preventing unnecessary computation while saving the optimal model weights.

10. LLM Integration & Streamlit Interface

10.1 Architecture Overview

After classification, the predicted intent and category labels — along with their confidence scores — are passed as structured context to the LLM. This bridges the gap between structured ML output and natural language conversation, allowing the LLM to generate contextually appropriate customer service responses without needing to interpret raw user text from scratch.

10.2 LLM Configuration

Parameter	Value / Description
Model	LLaMA 3.3 70B Versatile
Provider	Groq Cloud API
Temperature	0.3 (controlled, professional responses)
System Prompt	Structured prompt embedding classifier output as context
Persona	SupportBotV1 — professional female customer support agent
Memory	Full conversation history maintained per session
Confidence Logic	If overall confidence >= 0.60: direct response. If < 0.60: ask clarifying question

10.3 Chatbot Module

The Chatbot.py module handles the full conversational loop. On the first user message, both classifier models run inference to produce predictions, which are then locked into the system prompt for the entire session. Subsequent turns use the Groq API with full conversation history to maintain coherent multi-turn dialogue

10.5 Streamlit Interface

The Streamlit application provides the user-facing interface. When a user submits their problem description, the application simultaneously displays the predicted intent, category, and confidence scores in a visual dashboard, and automatically triggers the LLM with the locked classification context to begin the support conversation.

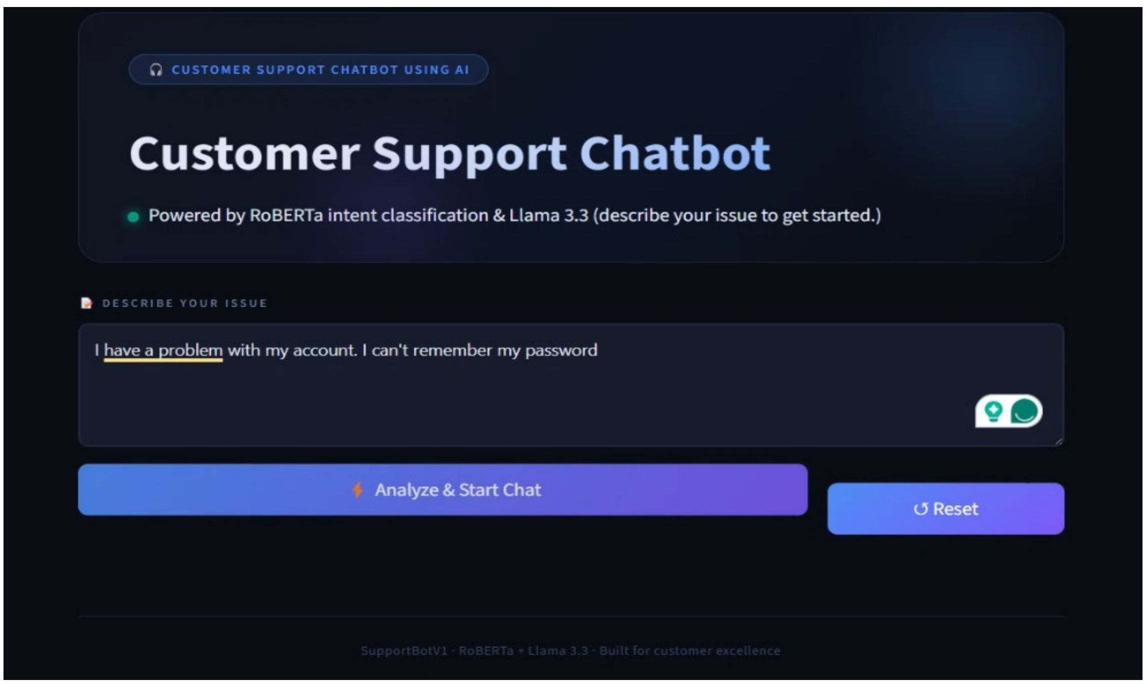


Figure 9: Streamlit UI — intent/category prediction panel with confidence visualization

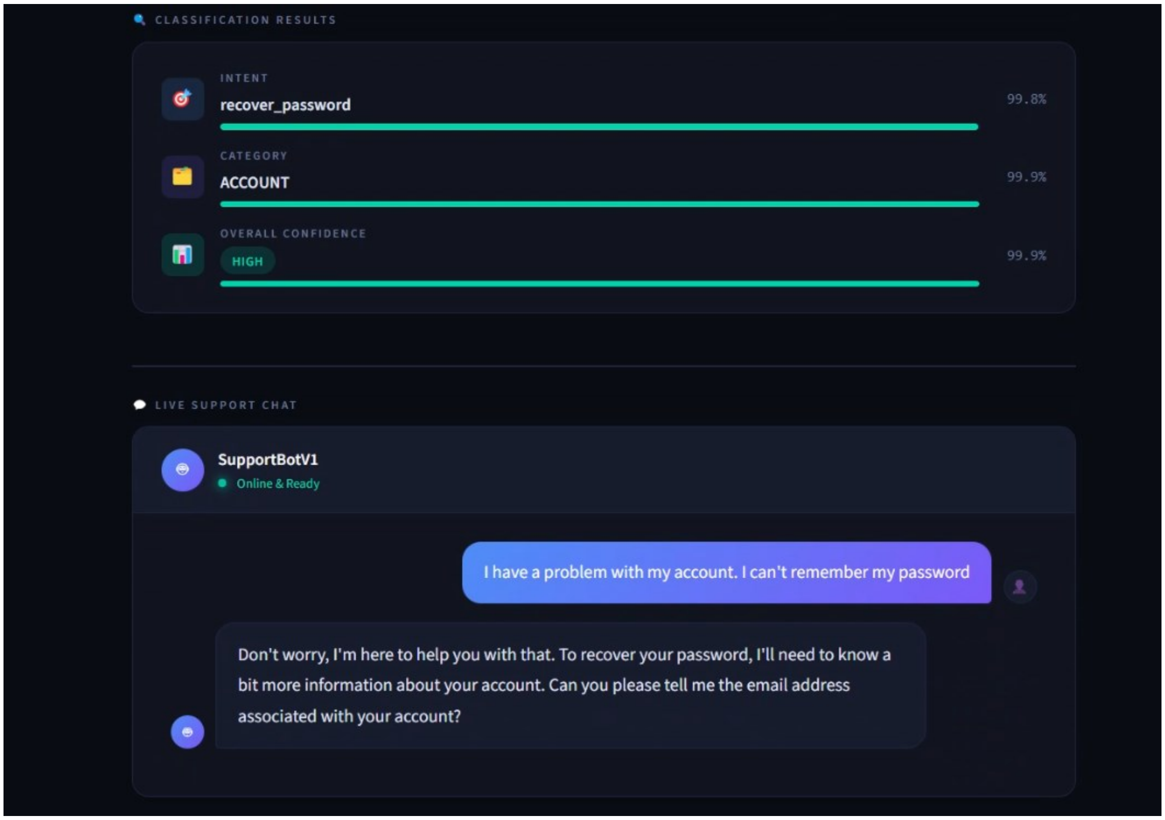


Figure 10: Streamlit UI — interactive chat interface powered by LLaMA 3.3 70B

---

11. System Workflow Diagram

The diagram below illustrates the complete end-to-end pipeline from raw user input through classification to LLM-powered response generation.

User Input Customer describes problem in natural language	→	**Text Cleaning** Minimal cleaning: URL/HTML removal, whitespace normalization	→	**RoBERTa Classifier #1** Intent model (27 classes) Predicts fine-grained intent
		↓ ↓		
RoBERTa Classifier #2 Category model (11 classes) Predicts high-level topic	→	**Confidence Routing** If score >= 0.60: direct reply If < 0.60: clarifying question	→	**LLaMA 3.3 70B (Groq Cloud)** Generates professional customer service response

Figure 11: End-to-end system workflow — from user input through dual classification to LLM-powered chat resolution

12. Conclusion

This project demonstrates a complete, production-ready customer support automation pipeline that combines state-of-the-art NLP classification with conversational AI capabilities. The system achieves the following:

12.1 Key Achievements

Parameter	Value / Description
Classification Accuracy	99.98% on the held-out test set (4,031 samples)
Macro F1 Score	0.9996 — consistent performance across all 11 categories
Single Misclassification	Only 1 error in 4,031 test predictions (0.02% error rate)
Training Speed	Converged in 4 epochs (~5 minutes on GPU)
Dual Model System	Simultaneous intent (27 classes) and category (11 classes) prediction
Zero-Code Interface	Streamlit UI requires no technical knowledge from the end user

12.2 Technical Contributions

- Demonstrated that RoBERTa-base with class-weighted loss and dynamic padding achieves near-perfect accuracy on the Bitext customer support dataset a strong baseline for production deployment.
- Designed a dual-model inference pipeline that runs both classifiers simultaneously, providing richer routing context (intent + category) to the downstream LLM.
- Implemented a stateful conversation system where the initial classification is locked into the LLM context, preventing topic drift and maintaining focus on the original customer issue throughout the session.